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ALICE L WALTON SCHOOL OF MEDICINE

Blood Pressure Control: Foundations and Frameworks

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Adapted from: Blood Pressure Control: Foundations and Frameworks

Listen to this Brick
0:00 / 25:45 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Describe the importance of blood pressure regulation.
2. Explain how systemic vascular resistance and cardiac output affect blood pressure.
3. Explain how t[illegible] renal and cardiovascul[illegible] pressure.

CASE CONNECTION

JB is an 82-year-old female with hypertension who is in your office for a routine follow-up visit. Her physical exam is unremarkable. When you ask JB if she has any questions or concerns, she says, "I was watching the news a few days ago, and the TV doctor said everyone's blood pressure should be 120/70 mm Hg. I'm on 10 mg of lisinopril, so I guess we need to see if it has me at the right level. I really don't want to take any more pills, though, to tell you the truth. They cost too much and I'm on a fixed income." You take her blood pressure (BP), which is 136/78 mm Hg.

How will you explain to JB that her blood pressure is meeting an acceptable target for someone her age? Consider your answers as you read, and we'll revisit JB at the end of the brick.

GO TO CONCLUSION ↓

Why Is Blood Pressure Regulation Important (LO1)?

Blood pressure (BP) regulation is critical. Too low, and the organs are not perfused with blood. We may become dizzy and then have syncope (passing out), kidney failure, blue toes, and eventually, shock and cell death. Too high, and there is an increased risk for heart attack, stroke, and kidney failure. How the body maintains a normal BP is an intricate balance of neural, hormonal, and physical pathways. Blood pressure can be reviewed in this hemodynamics brick. Here, we will fo[illegible]ute-to-minute basis.

CLINICAL CORRELATION

While you are taking blood pressure manually, the first sound heard (Korotkoff sounds) as you lower the pressure in the cuff relates to the systolic BP pulsing through the artery. The diastolic BP is the lower pressure when the pulsations become inaudible as the cuff deflates even more to not construct the artery.

What Are the Primary Factors That Regulate Blood Pressure (LO2)?

So why would we need to regulate the BP? The reasons range from simple challenges, like standing up from a chair, to bigger problems, like hemorrhage or dropping BP during an allergic reaction (anaphylaxis). Two main factors work together to regulate the BP in these situations: **systemic vascular resistance** and **cardiac output**. Each of these factors is under the integrated control of the body's physiologic systems, as we will discuss in the next section. First let's look at each of these two factors in more detail.

Systemic Vascular Resistance (SVR)

The systemic vascular resistance (SVR) is a key determinant of blood pressure. The SVR is related to the arterial blood vessels radius, or how contracted or dilated they are. The SVR is the **resistance to blood flow by all the combined systemic vessels.** (SVR is sometimes termed total peripheral resistance or TPR).

- Increasing SVR makes it more difficult for blood to flow from the larger arteries to the smaller ones, thereby increasing BP.
- Decreasing SVR allows blood to flow more easily through the larger arteries, thereby decreasing BP.

The constricting of blood vessels causes them to narrow and is called vasoconstriction (Figure 1); vasoconstriction increases the SVR and the blood pressure. In contrast, the relaxation of blood vessels decreases the SVR and lowers the blood pressure; this is called vasodilation.

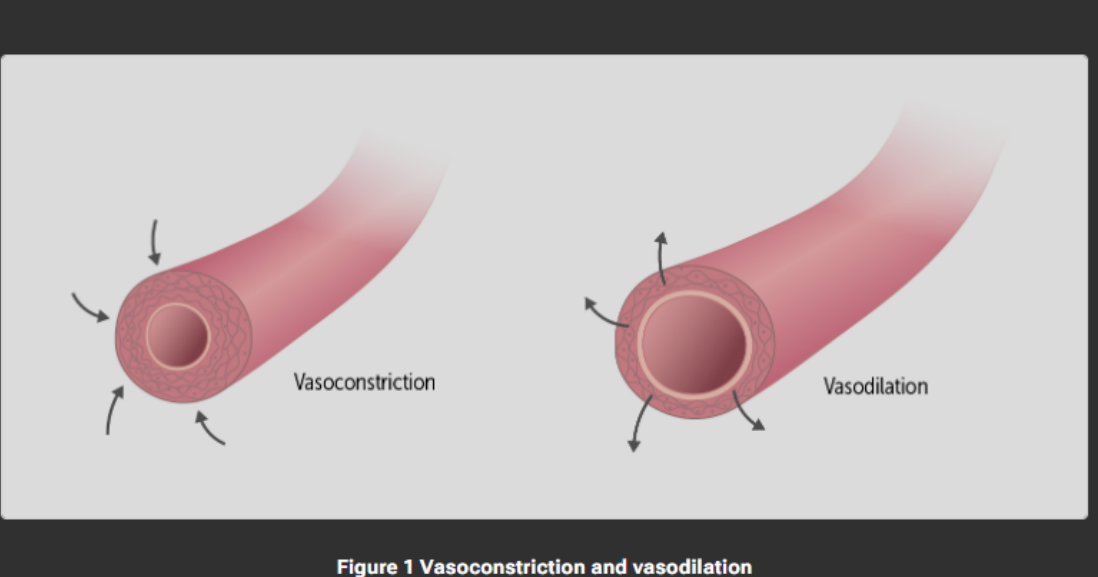

Figure 1 Vasoconstriction and vasodilation

The SVR is regulated by various factors, including autonomic nervous system input, vessel stretch (myogenic reflex), local factors (e.g., nitric oxide, oxygen, CO2), and circulating signaling molecules (e.g., vasopressin, endothelin, Ang II). All these factors l[illegible]uscles within the walls of the blood vessels contract or relax.

CLINICAL CORRELATION

In **septic shock**, bacteria release molecules into the circulation that cause arteriolar dilation and lower the SVR, thereby reducing the BP. The heart is unable to compensate, and fatal hypotension and multiorgan failure may result.

What effect would vasoconstriction and vasodilation have on BP, respectively?

CLINICAL CORRELATION

In **hypovolemic sho**[illegible]s from a hemorrhage. The body responds by inducing vasoconstriction, increasing cardiac contractility, and increasing heart rate; however, if these mechanisms are not enough to compensate for the volume loss, severe hypotension and organ failure can result.

Cardiac Output

Cardiac output (CO) is the other main factor in blood pressure regulation (see Cardiac- Compliance, contractility and preload brick). Recall that CO is the amount of blood the heart pumps per unit of time, and is dependent on the heart rate (HR, the number of heartbeats per minute) and stroke volume (SV, the amount of blood the heart pumps with each beat):

$$CO = HR \times SV$$

Several factors alter CO, and have been discussed elsewhere. However, below we relate these factors to the regulation of blood pressure:

- **Heart rate (HR):** A higher HR, the greater the CO and more blood is pumped by the heart. More blood being pumped results in higher pressure, making CO (and, therefore, HR and SV) crucial in how the body regulates BP.
- **Preload:** This is the stretch of the left ventricle, and the main determinant of preload is the[illegible]d volume is largely determined by your fluid intake and by your kidneys' ability to retain fluid. Normally, there are about 4.7-5.5 L of blood in an average human body. If that volume decreases, for example, due to dehydration in the desert or hemorrhaging after an accident, there may be insufficient blood filling the cardiovascular system to maintain preload, resulting in a fall in cardiac output and blood pressure.
- **Contractility or inotropy** is the ability of the myocardial muscle to generate force, and as such, it increases stroke volume (SV) and blood pressure. Contractility can be increased by nervous input (sympathetic nerves), but it falls after a heart attack (myocardial infarction) or other types of myocardial disease.
- **SVR, Regulation of blood vessel tone via circulating factors** like antidiuretic hormone (ADH, also called vasopressin), angiotensin II produced by the renin-angiotensin-aldosterone system RAAS, and nitric oxide can independently exert control over the cardiovascular system. They are released both under normal physiologic conditions and in disease states. Circulating factors play a crucial role in both short-term and long-term blood pressure (BP) control.

CLINICAL CORRELATION

Vasoconstriction will raise SVR, and thus the blood pressure, in normal patients. However, a high SVR in the setting of a failing heart may reduce cardiac output, which, in turn, could lower the blood pressure. Thus this compensatory mechanism can become maladaptive.

In **cardiogenic shock**, the CO drops because of myocardial injury (eg, infarction, toxic injury); periphe[illegible]t compensate adequately, leading to severe hypotension and organ failure.

How does preload affect cardiac output and blood pressure?

How Is Blood Pressure Regulation Integrated (LO3)?

The regulation of blood pressure is an interplay of cardiovascular, renal, and neural physiology. All three systems work together to ensure our BPs and circulating blood volume remain relatively stable (Figure 2).

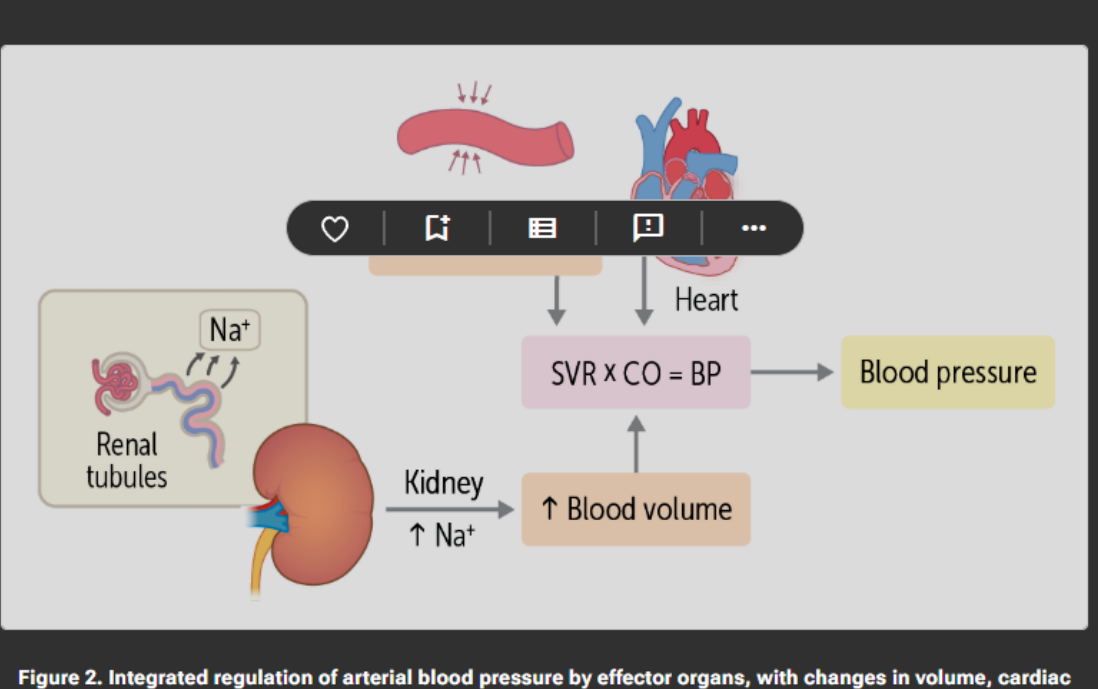

Figure 2. Integrated regulation of arterial blood pressure by effector organs, with changes in volume, cardiac output (CO), and resistance or vasoconstriction (SVR).

What factors influence cardiac output and blood pressure?

Figure 2 shows that interrelatedness of the the factors we've discussed: cardiac output (CO), systemi[illegible] pressure.

Note that in clinical situations, when one of these factors is increased or decreased, your BP does not necessarily change. This is because the other two factors can compensate. For example, if you develop heart failure after a heart attack (decreased contractility, decreased CO), the kidney will retain fluid to increase the blood volume and the blood vessels will constrict, raising the SVR. So, the BP may not actually fall unless the heart failure becomes very severe, in which case the compensations will not be able to keep up and the BP will fall below normal (hypotension).

The Autonomic Nervous System: Short–Term BP Control

The autonomic nervous system is important in the instant, second-to-second control of your blood pressure. How fast can this system respond? Consider a scenario where you accidentally submerge your hand in boiling water. Not only would it cause pain within fractions of a second, but the nerve endings in your hand would also sense the temperature and send signals to the central nervous system telling you to pull your hand away. This is how quickly a neural circuit can process information—a concept that holds true for regulation of BP and volume as well. Simply put, neural signals modulate acute changes in pressure and volume and are ultimately responsible for making sure neither one changes much on a second-by-second basis.

The hypothalamus is the processing center that coordinates the sensory inputs (eg, from the heart, blood vessels, and elsewhere) with the outputs (i.e., sympathetic and parasympathetic nerve signals to the heart, blood vessels, and kidneys). How do nerves sense changes in blood pressure or volume? By detecting change in the stretch of arterial vasculature:

- Too much stretc[illegible]pressure.
- Too little stretch means either the pressure or the volume is not enough.

This is the basis of the **baroreceptor reflex** (see brick). Briefly, baroreceptors in the carotid artery and aorta sense changes in arterial stretch. Nerve signals are then sent to the medulla via cranial nerves IX and X for processing. The autonomic sympathetic and parasympathetic nervous systems send signals back to the heart, kidneys, and blood vessels to make the appropriate compensatory changes to bring the BP back toward normal. These signals can control each of the three determinants of BP that we have discussed: CO, SVR, and blood volume. For example, in response to low blood pressure, some of these efferent signal responses include increases in:

- HR, which increases the CO
- Cardiac contractility, which increases the CO
- Renal salt and water reabsorption, which increases the blood volume
- Muscle tone in the arterioles, which increases SVR

How does the autonomic nervous system (ANS) detect changes in blood pressure or volume?

The Kidneys: Long–Term BP Control

The kidneys are the primary regulators of blood volume. While neural mechanisms modulate acute, short-term changes in BP and volume, the kidneys are responsible for regulating BP changes over the long term, on the order of days to weeks. Why is this? Urination removes fluid from the body, and this fluid that is ultimately derived from circulating blood volume:

- You urinate less (decreased output) when your blood volume or pressure is low to conserve volume.
- You urinate more (increased output) when your blood volume or pressure is high to reduce blood volume.

This should inherently make sense if we consider situations when we may have elevated or decreased fluid or BP. For example, if we were dehydrated and wandering in the desert with an empty canteen, we wouldn't want to lose fluid as urine: we'd want to reabsorb the fluid to maintain our volume. Likewise, if we drink too much fluid, we need to urinate it out or risk having the fluid buildup in our blood vessels and tissues (edema).

CLINICAL CORRELATION

Patients with **end-stage kidney disease** have failing kidneys and reduced urine output, and thus they can no longer control their blood volume by urination. These patients develop severe edema and high BP due to the very high blood volume, and they must rely on regular dialysis treatment to remove their excess fluid, at least until they can g[illegible]

The kidneys regulate urine output by adjusting the excretion of sodium chloride (salt) and water. Both are important because, in most tissues, movement of salt causes the movement of water via osmotic pressure (water follows solute). There are many mechanisms that promote water and salt reabsorption in the kidney, including the sympathetic nervous system and the hormones aldosterone, angiotensin II, and antidiuretic hormone (ADH or vasopressin).

Likewise, there are hormones that promote salt excretion in the kidney (e.g., natriuretic peptides that are released when intravascular volume is too high). Taken together, all these integrate with the neural and cardiovascular systems to maintain our blood pressure.

How do the kidneys help regulate long-term blood volume?

The major mediator of blood volume and osmolarity control is the **renin-angiotensin-aldosterone system** (RAAS). The RAAS revs up when there is low blood volume or low blood pressure, stimulating the kidneys to reabsorb sodium from the nephrons. Ang II produced during the RAAS have several effects to control volume and osmolarity, with effectors listed in Figure 3. Because water generally follows sodium into cells, sodium reabsorption also leads to water reabsorption, helping to correct the low blood volume.

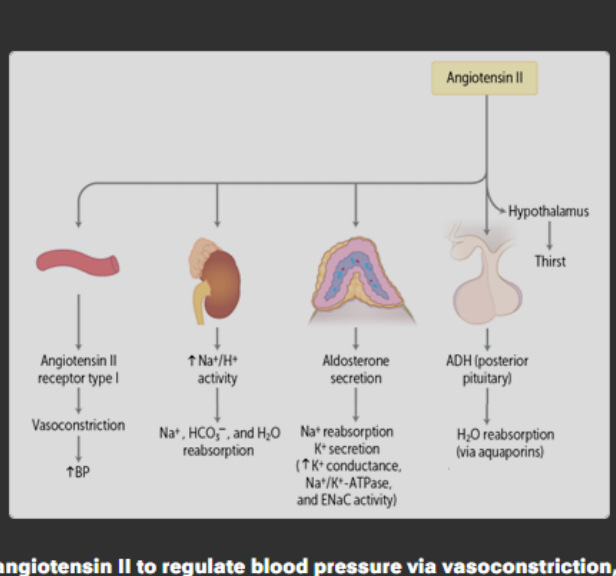
Angiotensin II
Hypothalamus
Thirst
Angiotensin II receptor type I
Vasoconstriction
↑BP
↑Na⁺/H⁺ activity
Na⁺, HCO₃⁻, and H₂O reabsorption
Aldosterone secretion
Na⁺ reabsorption
K⁺ secretion
(↑K⁺ conductance, Na⁺/K⁺-ATPase, and ENaC activity)
ADH (posterior pituitary)
H₂O reabsorption (via aquaporins)

Figure 3 Effects of angiotensin II to regulate blood pressure via vasoconstriction, and salt and water reabsorption

How does the renin-angiotensin-aldosterone system (RAAS) regulate blood pressure?

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to JB, how will you assure her that her blood pressure is in an acceptable range for someone her age?

You tell JB that many factors affect blood pressure, such as aging, stress, and lifestyle choices. As people age, their blood pressure tends to be higher because of calcium buildup within the arteries, making them stiffer and less compliant. "The pipes get a little stiffer," you tell her. "The heart has to work a little harder to deliver oxygen-rich blood through your body." A blood pressure of 136/78 mm Hg is right on target for an otherwise healthy 82-year-old female. You encourage JB to continue taking her medications, eating a low-sodium diet, and following up with her appointments. She is relieved she doesn't have to take more medication.

Summary

- Blood pressure re[illegible]organ perfusion and preventing conditi[illegible]ck.
- Systemic vascular resistance (SVR) and cardiac output are the primary factors regulating blood pressure, influenced by neural, hormonal, and physical pathways.
- Vasoconstriction increases SVR and blood pressure, while vasodilation decreases SVR and blood pressure.
- Systemic vascular resistance is determined by the nervous system and circulating humoral factors.
- Cardiac output is determined by the heart rate and the stroke volume; it is regulated by cardiac contractility, the sympathetic nervous system, renal system, humoral factors, blood volume, and blood pressure.
- The autonomic nervous system provides short-term blood pressure control through baroreceptor reflexes, affecting heart rate, contractility, and blood vessel tone.
- The kidneys regulate long-term blood pressure by adjusting sodium and water excretion, primarily through the renin-angiotensin-aldosterone system (RAAS).

Review Questions

1. A 23-year-old male is brought into the emergency department with a gunshot wound to the abdomen. An emergency medical technician is applying pressure to the wound and it is not currently bleeding. The patient's vital signs are immediately taken: his blood pressure is 100/80 mm Hg and pulse is 150/min. He is confused and appears very pale. Which of the following best explains how his body is compensating to maintain mean arterial pressure (MAP)?

A. ○ Decreasi[illegible]
B. ○ Decreasing venous return
C. ○ Increasing cardiac output
D. ● Increasing systemic vascular resistance
E. ○ Lowering heart rate

Hide Explanation

Correct answer is D). Increasing systemic vascular resistance. Mean arterial pressure (MAP) is determined by the equation:

$$MAP = CO \times SVR$$

In hypovolemic shock due to blood loss, the venous return and stroke volume is reduced. To compensate, the sympathetic nervous system activates mechanisms to maintain MAP, including vasoconstriction (increasing SVR) and tachycardia (increasing HR to compensate for reduced stroke volume). Incorrect answers: A). Decreasing systemic vascular resistance would lower MAP further, worsening hypotension. B). Decreasing venous return would reduce preload, further decreasing cardiac output and MAP. C). Increasing cardiac output is incorrect because stroke volume is significantly reduced in hypovolemia; heart rate increases, but CO is still impaired. E). Lowering heart rate would decrease cardiac output and further reduce MAP, worsening shock.

2. A 25-year-old man suddenly stands up from a seated position after sitting for an extended period. He experiences a brief episode of dizziness but quickly recovers. Which of[illegible]s is most responsible for stabilizing his blood pressure?

A. ● Baroreceptor reflex
B. ○ Decreased cardiac contractility
C. ○ Decreased sympathetic activity
D. ○ Increased vagal tone

Hide Explanation

Correct answer is A). When standing up, blood pools in the lower body, lowering venous return and cardiac output. Baroreceptors detect this and activate the sympathetic nervous system, increasing heart rate and vasoconstriction to restore blood pressure. Incorrect answers: B). Decreased contractility would reduce cardiac output and worsen hypotension. C). Decreased sympathetic activity would lead to vasodilation and lower BP. D). Increased vagal tone slows the heart and lowers BP, the opposite of the needed response.

3. A 76-year-old woman with a history of hypertension and chronic kidney disease presents to the emergency department after experiencing dizziness and confusion. She reports that she has had nausea and vomiting for the past three days, leading to decreased oral water intake. On examination, her blood pressure is 85/55 mm Hg, and her heart rate is 110/min. Laboratory results reveal elevated blood urea nitrogen (BUN) and creatinine levels, indicating decreased urine ou[illegible]ollowing best explains this patients symptoms?

A. ○ Decreased sympathetic nervous system activity
B. ● Decreased renal perfusion due to hypotension
C. ○ Increased stroke volume due to dehydration
D. ○ Increased venous return due to decreased blood volume

Hide Explanation

Correct answer is B). Decreased renal perfusion due to hypotension. This patient is experiencing hypovolemia due to nausea, vomiting, and poor oral intake, leading to low blood pressure and reduced organ perfusion. The kidneys are particularly sensitive to hypoperfusion, as evidenced by her elevated BUN and creatinine, indicating prerenal azotemia (kidneys fail to filter waste products). Blood pressure regulation is essential to ensure adequate perfusion of vital organs, including the kidneys and brain, which explains her dizziness and confusion. Incorrect answers: A). Hypotension activates the sympathetic nervous system, leading to increased heart rate and vasoconstriction to compensate for low blood pressure. C). Dehydration decreases stroke volume due to reduced preload. D). A lower blood volume reduces venous return over the long term, impairing cardiac output.

4. A 55-year-old female comes to the emergency department following a motor vehicle accident, with blood loss. Her vital signs: pulse 120 beats/min, blood pressure 110/70 mm Hg. Which of the following is the most likely change that would be observed[illegible]

A. ○ Higher CO, lower SVR
B. ○ Higher CO, unchanged SVR
C. ○ Lower CO, lower SVR
D. ● Lower CO, higher SVR
E. ○ Unchanged CO, lower SVR

Hide Explanation

The correct answer is D). Blood loss from the venous system would initially cause a drop in the venous return to the heart, lowering the cardiac output (CO) and subsequently the blood pressure. The reduction in volume would immediately be picked up by the baroreceptors in the carotid artery and aorta. The baroreceptors would then send signals to the brain, which would process the information and send sympathetic nervous signals to the heart and vessels. The heart rate and contractility would increase, offsetting the initial drop in CO (but likely lower), and the arterioles would constrict, increasing the systemic vascular resistance (SVR). Both changes aim to bring the blood pressure back to normal, but do not get there entirely. Incorrect answers: A). C). and E). the SVR would not be lower or decrease; B). The SVR would also not remain unchanged,and the CO would not be higher with blood loss and low BP.

H[illegible]k?

References

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- Hemodynamic profiles of shock on pulmonary artery catheter in adults. UpToDate; 2022. Accessed September 15, 2022. https://www.uptodate.com/contents/image?imageKey=PULM%2F99516

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the importance of blood pressure regulation.~~

02 ☑ ~~Explain how systemic vascular resistance and cardiac output affect blood pressure.~~

03 ☑ ~~Explain how the autonomic nervous system and the renal and cardiovascular systems integrate to regulate blood pressure.~~

References

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Objective Confidence: 3/3